

Digital Signal Processing – Lecture 1

Introduction to SSE

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About Course

- 7 Sessions
- Emphasis on applying DSP using Python
- 50% Introduction to Smart Systems Engineering

Some Books

- Digital Signal Processing First
 - By James H. McClellan, Ronald W. Schafer, Mark A. Yoder
 - ISBN: 9781292113869
 - [https://mrce.in/ebooks/Digital%20Signal%20Processing%20\(DSP\)%20First%202nd%20Ed.pdf](https://mrce.in/ebooks/Digital%20Signal%20Processing%20(DSP)%20First%202nd%20Ed.pdf)
- Think DSP
 - By Allen B. Downey
 - ISBN: 9781491938515
 - <https://greenteapress.com/thinkdsp/thinkdsp.pdf>
- Engineering Mathematics
 - ISBN: 978-0273719779
 - <https://www.amazon.com/Engineering-Mathematics-4th-edn-Communications/dp/0273719777>

Today

- Discrete vs Continuous
- Linear Time Invariant Systems
- Discrete Time Unit Impulse
- Discrete Time Impulse Response
- Exercises

Where is DSP used?

Applications [\[edit \]](#)

General application areas for DSP include

- Audio signal processing
- Audio data compression e.g. MP3
- Video data compression
- Computer graphics
- Digital image processing
- Photo manipulation
- Speech processing
- Speech recognition
- Data transmission
- Radar
- Sonar
- Financial signal processing
- Economic forecasting
- Seismology
- Biomedicine
- Weather forecasting

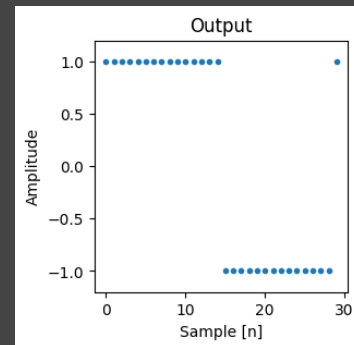
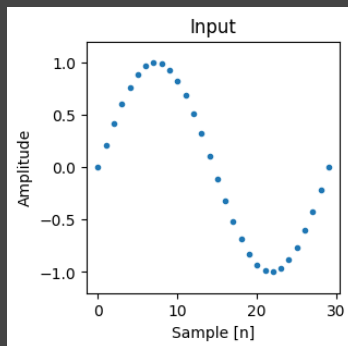
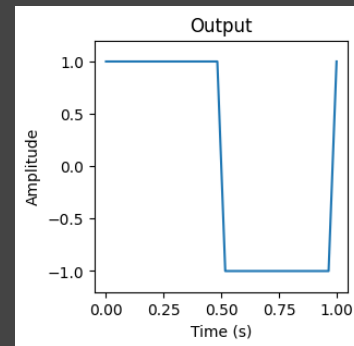
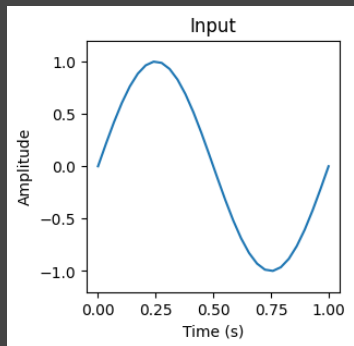
Specific examples include [speech coding](#) and transmission in digital [mobile phones](#), [room correction](#) of sound in [hi-fi](#) and [sound reinforcement](#) applications, analysis and control of [industrial processes](#), [medical imaging](#) such as [CAT](#) scans and [MRI](#), [audio crossovers](#) and [equalization](#), [digital synthesizers](#), and audio [effects units](#).^[22]

Linear Time Invariant Systems

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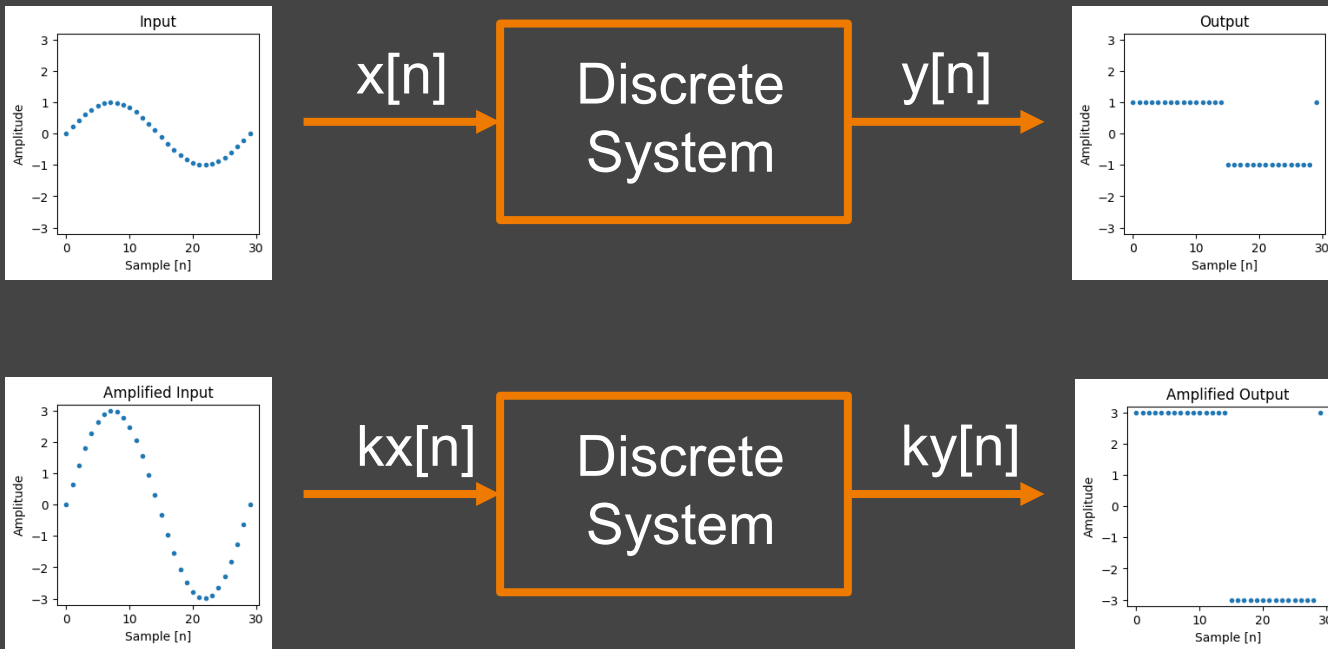
System



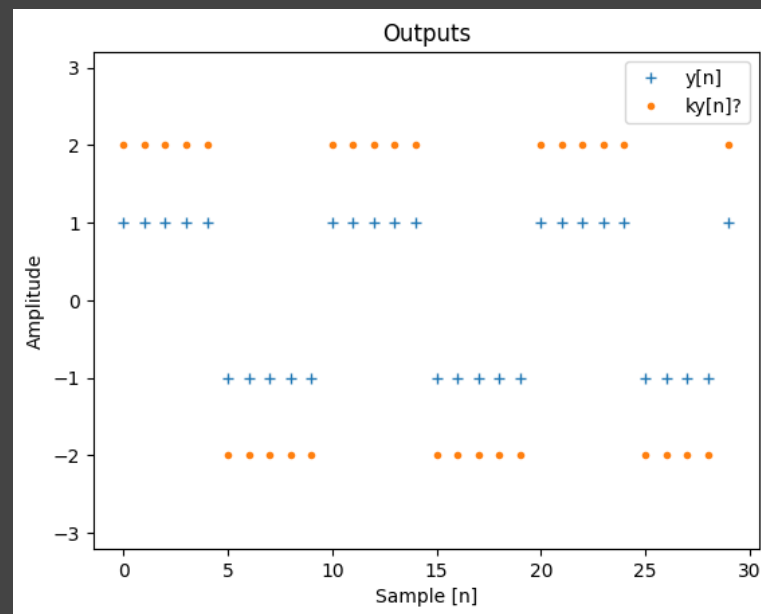
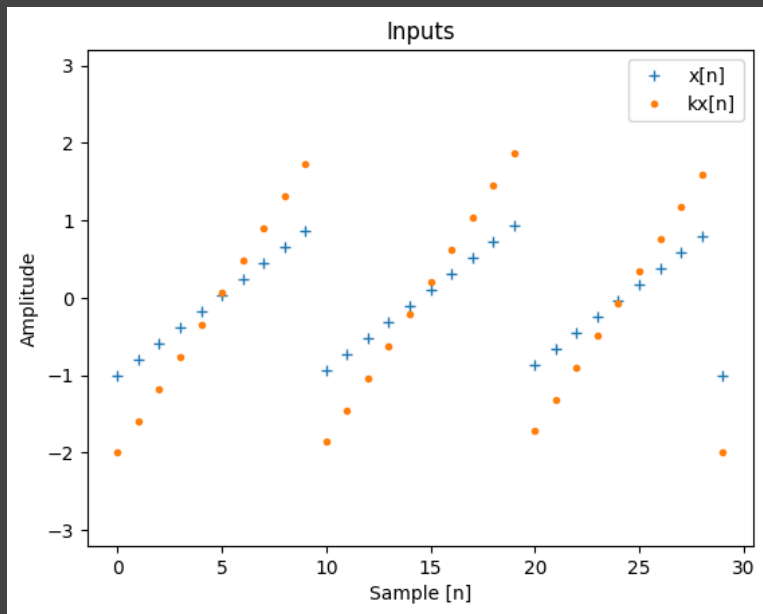
Linear Time Invariant System

- Homogeneous
- Additive
- Shift Invariance

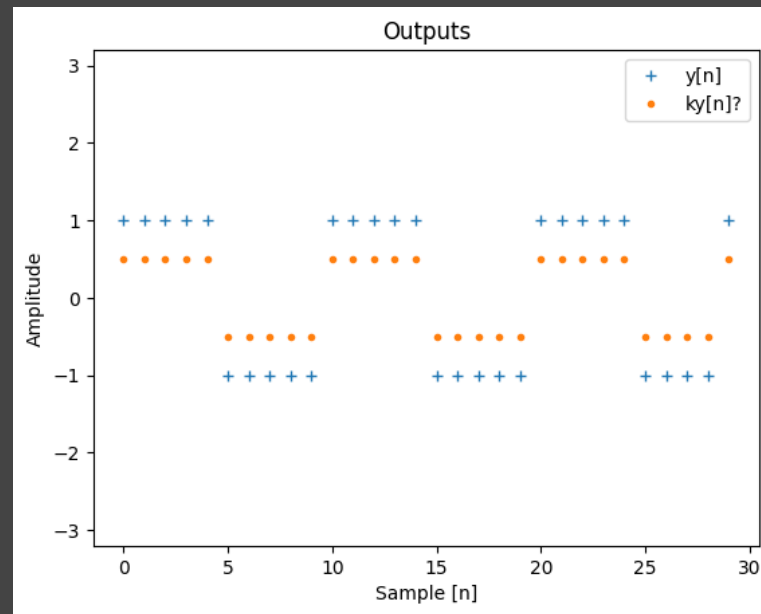
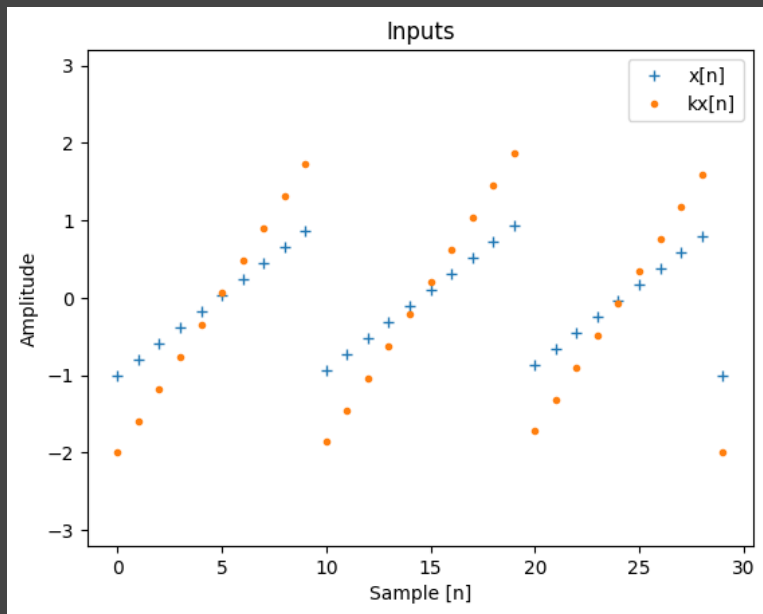
Homogeneous



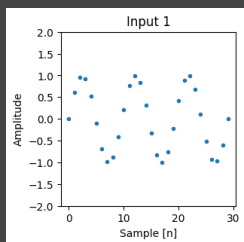
Homogeneous?



Homogeneous?



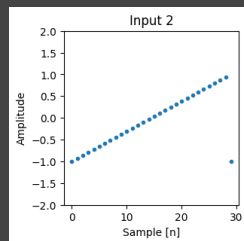
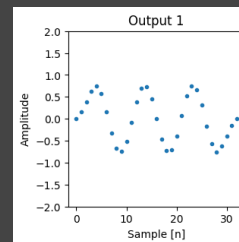
Additive



$x_1[n]$

Discrete
System

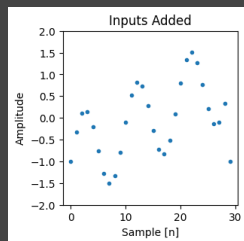
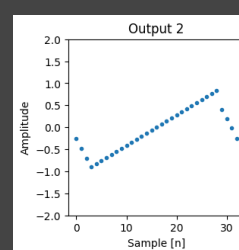
$y_1[n]$



$x_2[n]$

Discrete
System

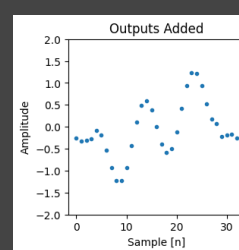
$y_2[n]$



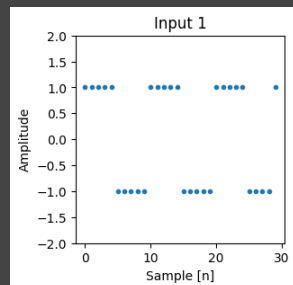
$x_1[n] + x_2[n]$

Discrete
System

$y_1[n] + y_2[n]$



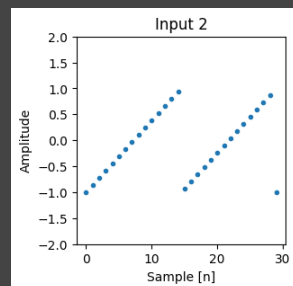
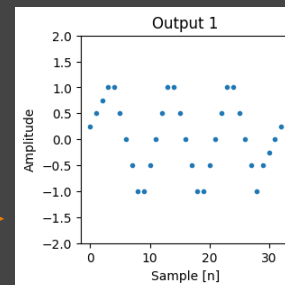
Additive?



$x_1[n]$

Discrete
System

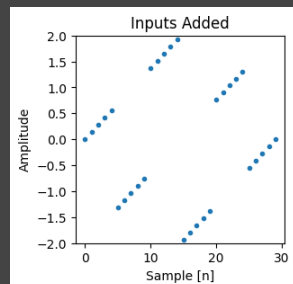
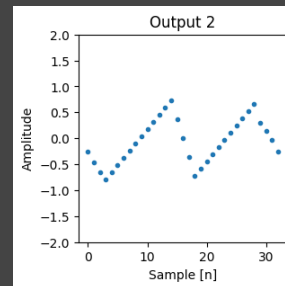
$y_1[n]$



$x_2[n]$

Discrete
System

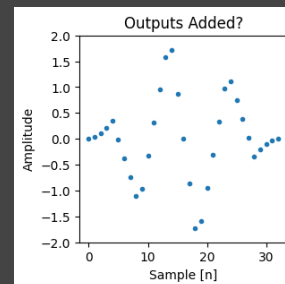
$y_2[n]$



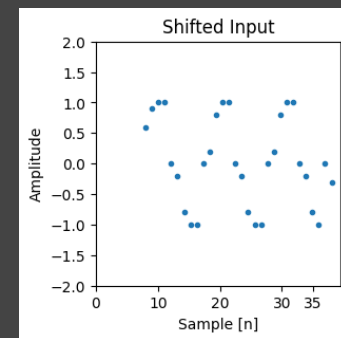
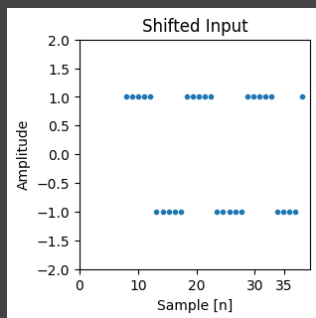
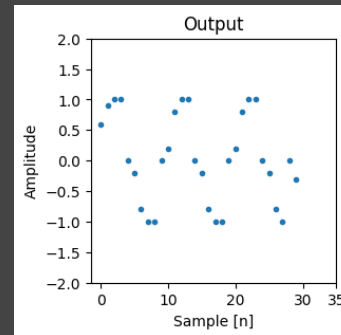
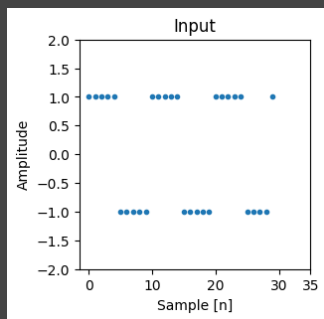
$x_1[n] + x_2[n]$

Discrete
System

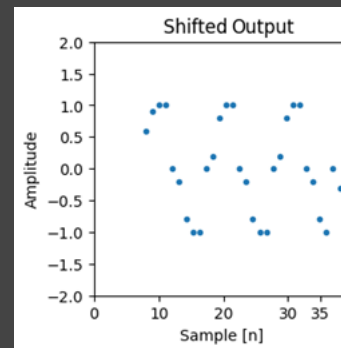
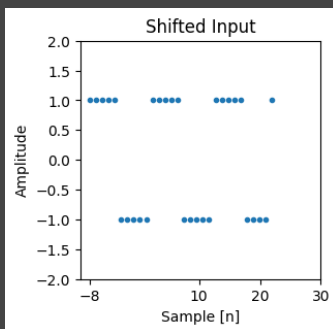
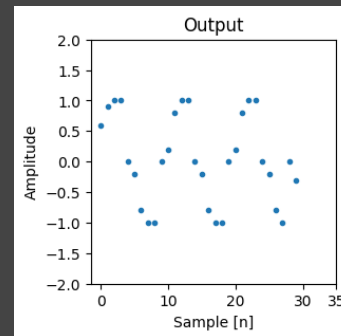
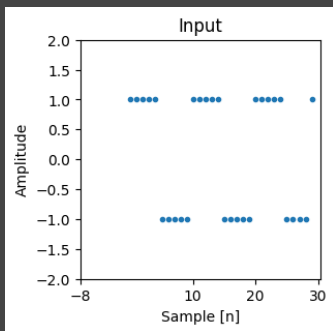
$y_1[n] + y_2[n]$



Shift Invariance



Shift Invariance?



LTI

- Homogeneous
- Additive
- Shift Invariance

Questions?

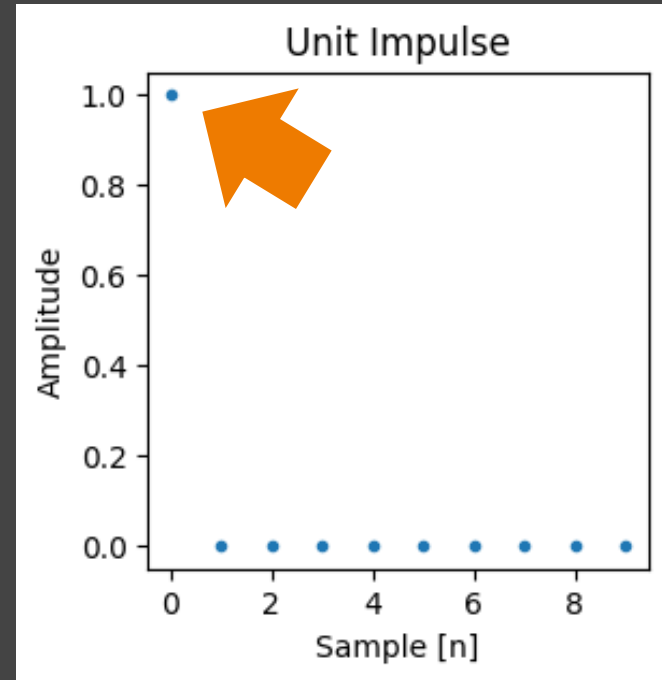
Discrete Time Unit Impulse

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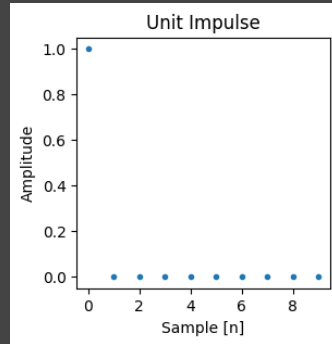
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Discrete Unit Impulse

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$



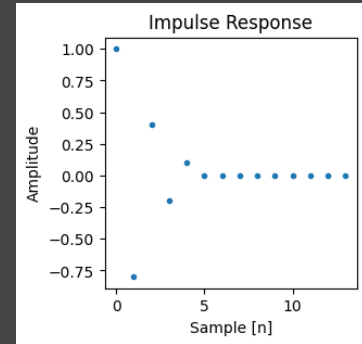
Impulse Response



$\delta[n]$

Discrete
System

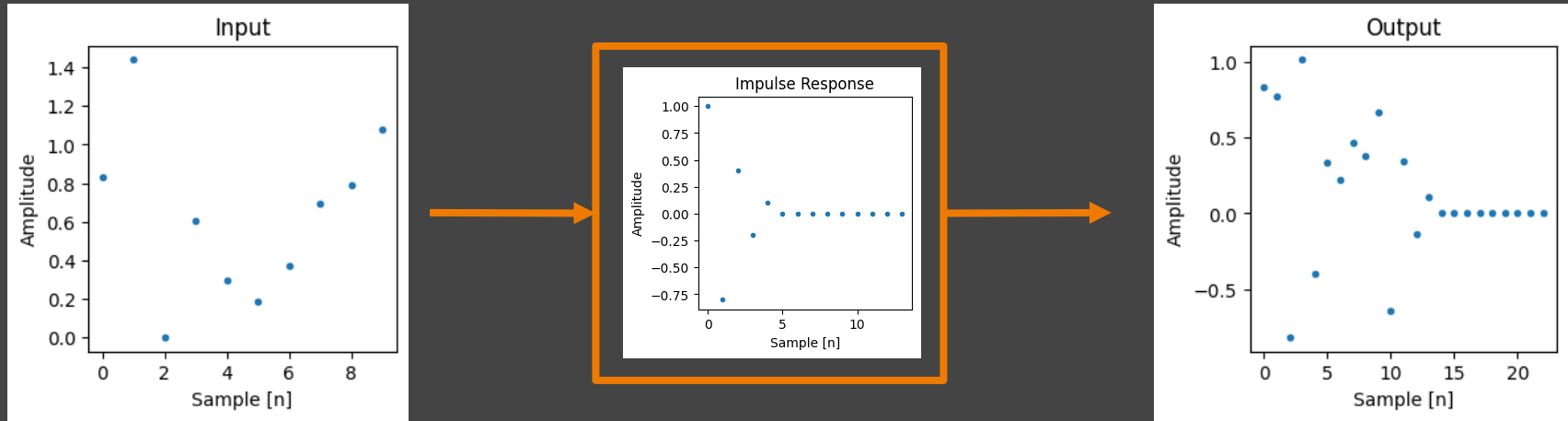
$h[n]$



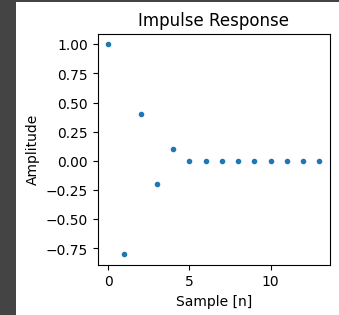
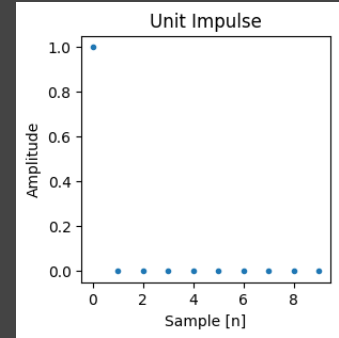
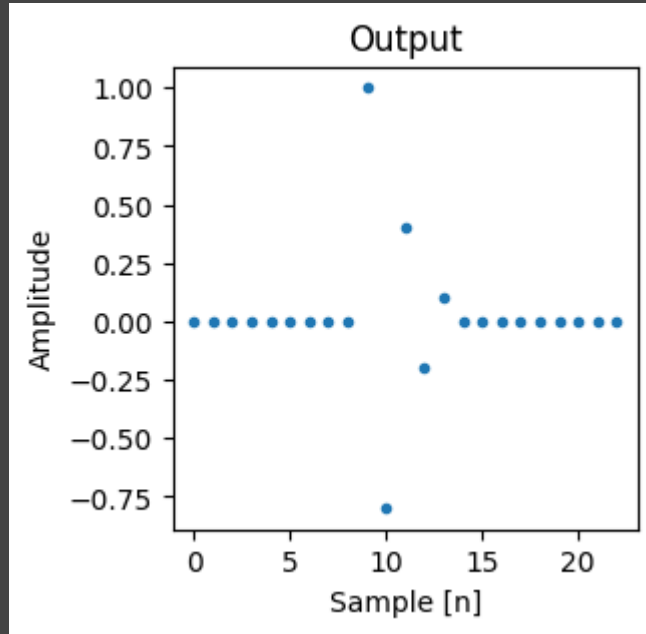
Discrete Time Impulse Response



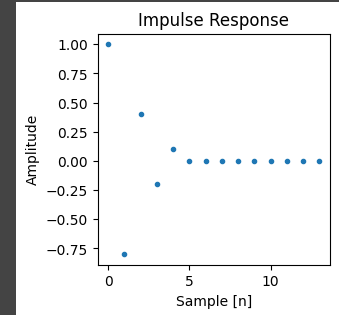
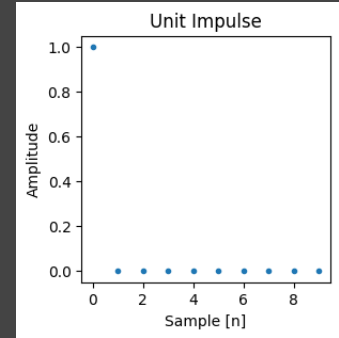
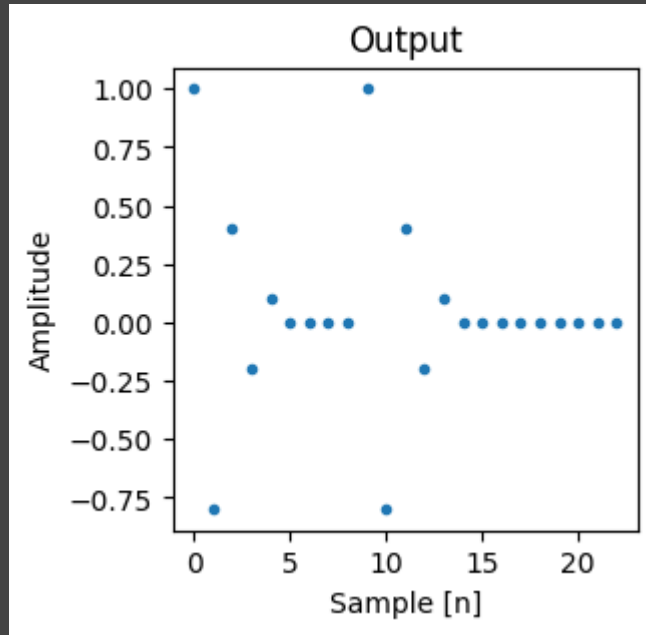
Discrete Time Impulse Response



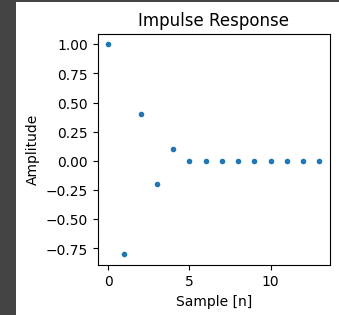
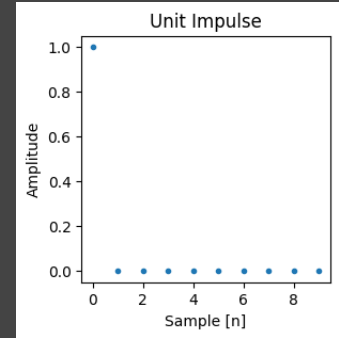
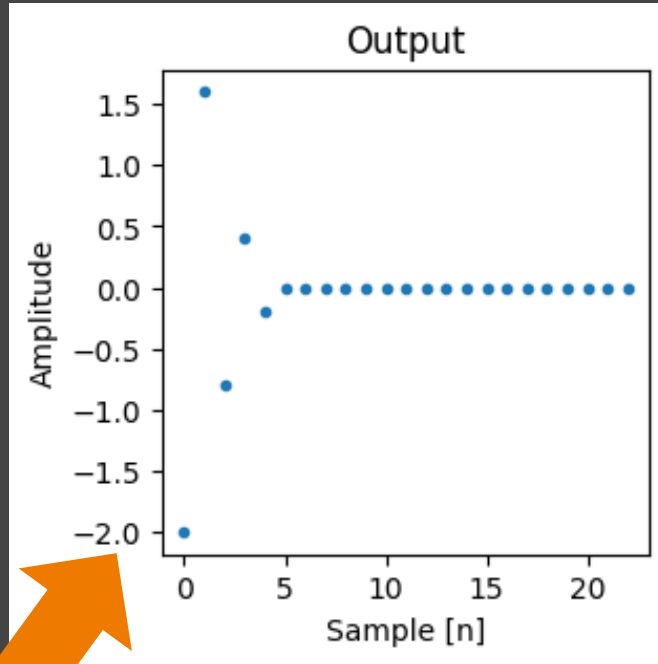
What input happened here?



What input happened here?



What input happened here?



Do not forget the exercises!